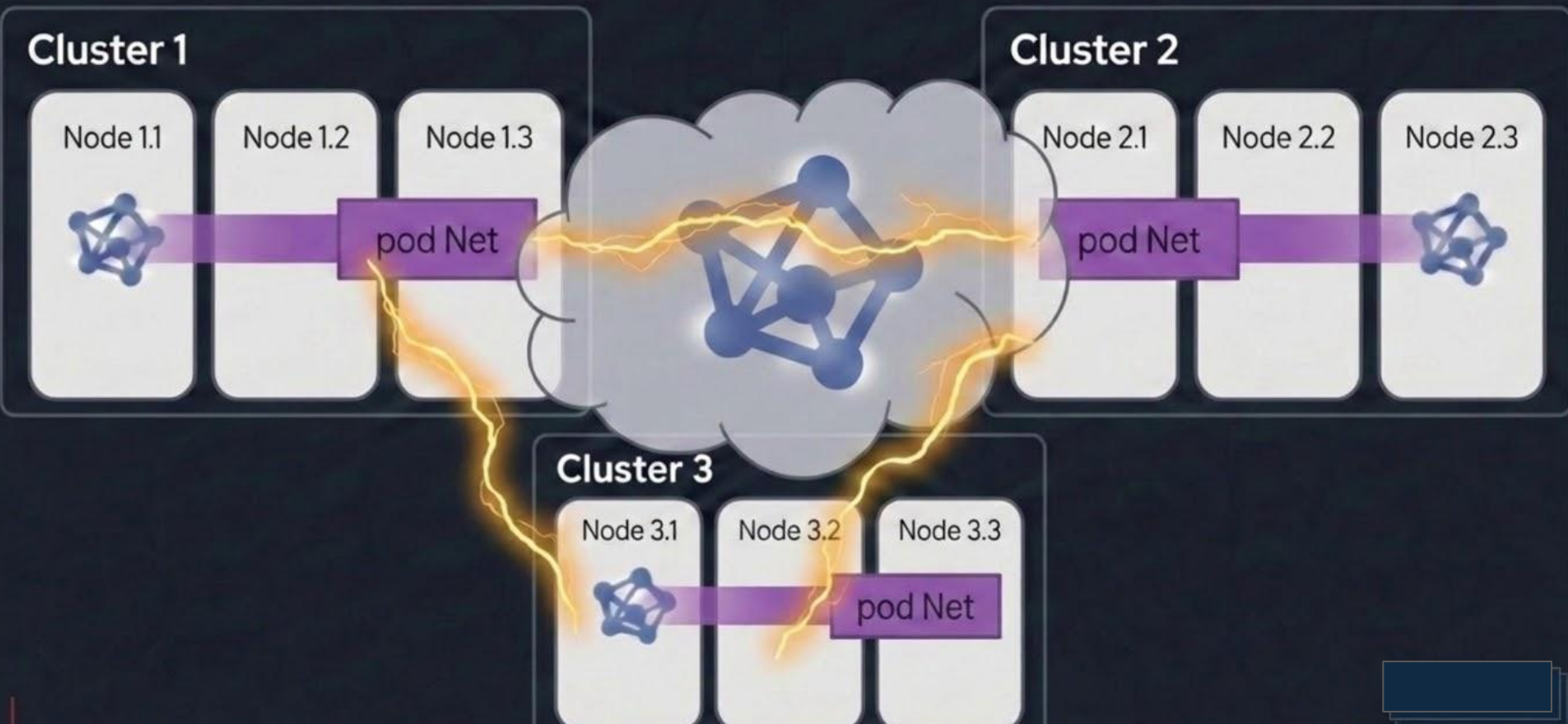


OVNK multi cluster networking

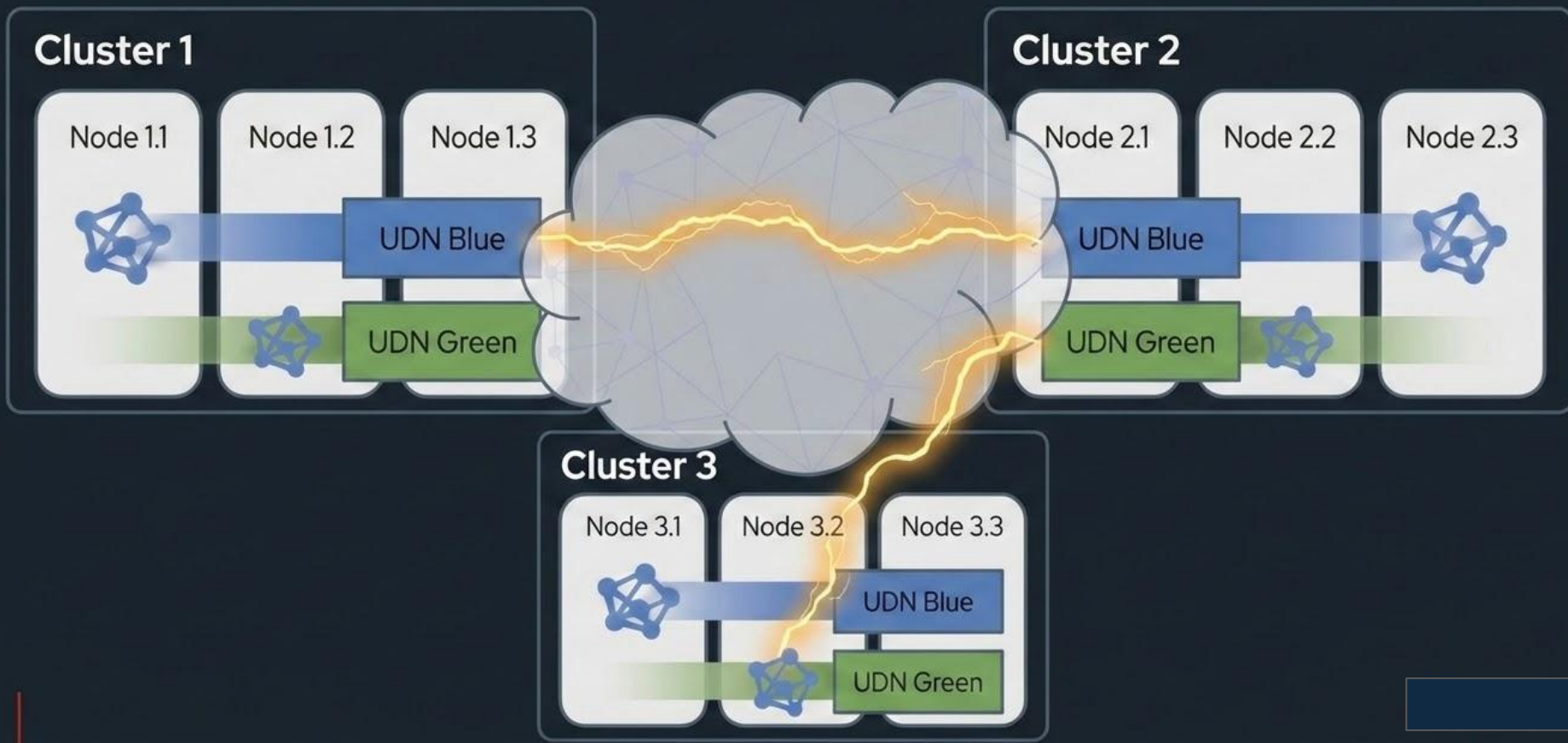
PoC

Yossi Boaron

Multi-Cluster pod Net Connectivity



Multi-Cluster UDN Connectivity



Multi-Cluster networking: some open questions

1. Underlay Technology

- EVPN: Industry standard, OVN-K already has support.
- Other Approaches

2. Multi-Cluster Aware CP

Make clusters aware of each other (sync nodes details, sync VNIs, ASNs, non-conflict VTEP CIDRs for EVPN).

- External component
- Extending OVN-K (something like Calico/Cilium mesh style)
- None , distributed manual config

3. Integration with OVN-K

- Extend OVN-K existing controllers.
- External component that will configure OVN-K APIs (VTEP, CUDN, RA).

4. API Design

- **BGP Bits:** BGP super users
- **Intent-Based:** Declarative (simplified). "I want to stretch Blue UDN from cluster1"

5. Supported Inter cluster network type

- trusted network
- Untrusted network

6. networks type to stretch

- Default pod network
- UDN

MCN Demo

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- 
- UDNs

MCN Agent

Key Features



Extend User Defined Networks across multiple clusters



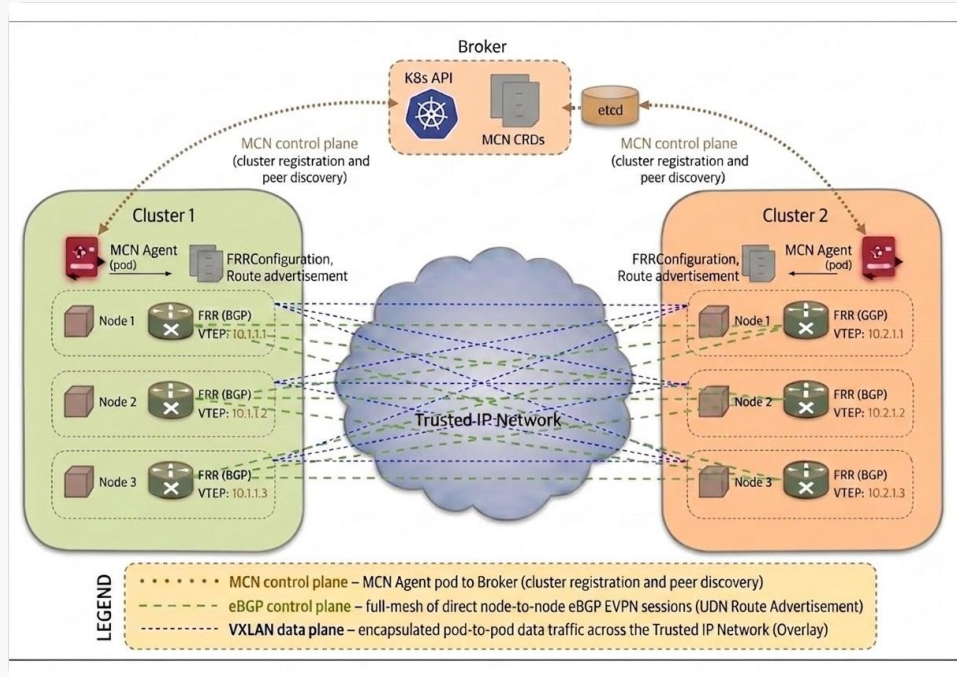
Managed clusters discovery and BGP peering



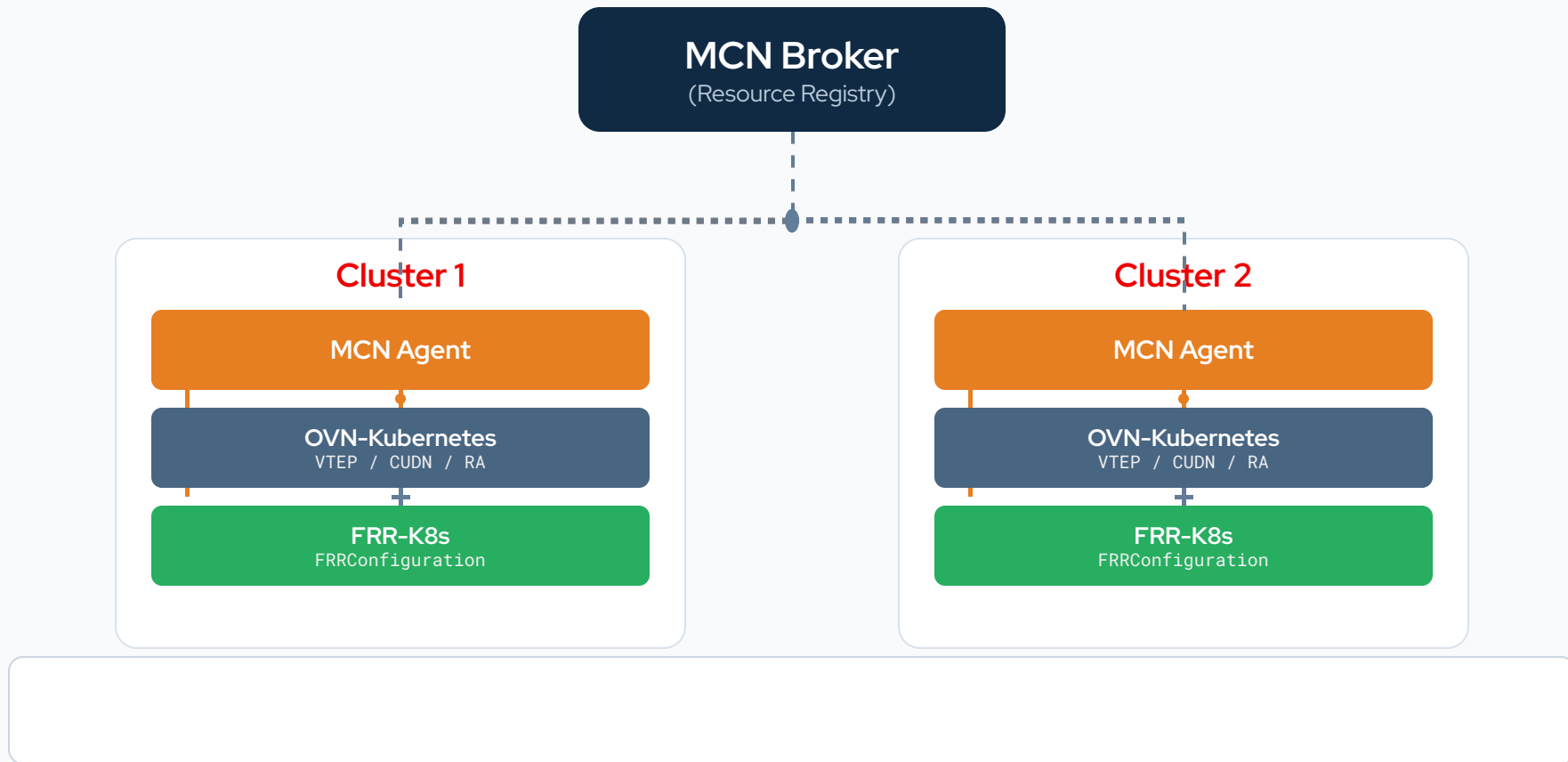
Conflict-free VTEP CIDR and ASN allocation



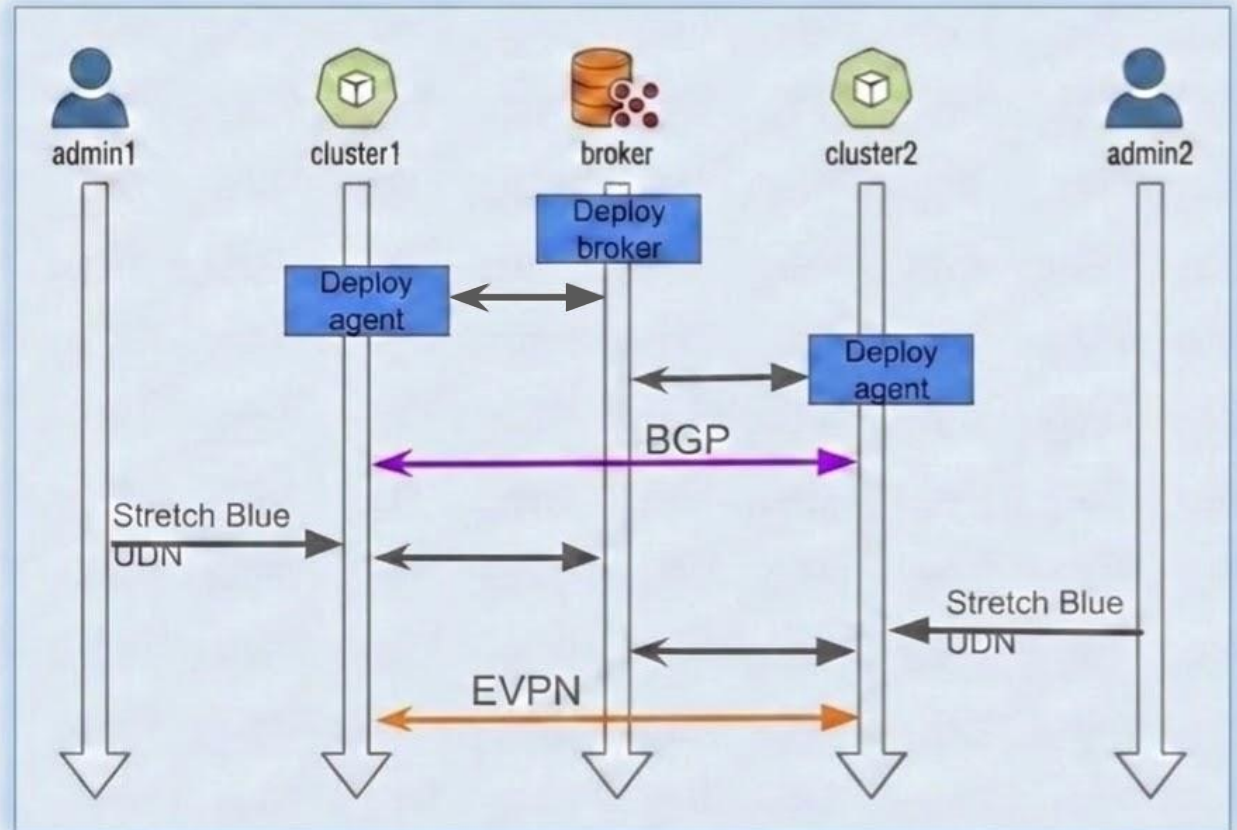
Declarative UDN connectivity across clusters



MCN: Multi-Cluster Layered Architecture



Sequence Diagram: MCN Interaction Flow



Manual vs MCN

Manual Configuration

01. Node Discovery & BGP

- Collect node IPs and sync across clusters
- Manual generation of FRRConfiguration per node

02. VTEP CIDR Allocation

- Allocate CIDR per cluster
- Verify non overlapping CIDRs

03. VNI/RT Assignment

- Manual allocation
- Cross-cluster value sync

04. Stretch UDN

- Build EVPN & RouteAdv spec
- Repeat matching config on all sites

05. Stretch Default pod Net

- Create RouteAdv CR on each cluster

06. BGP/EVPN control

- Full control
- all BGP/EVPN parameters configurable

MCN

01. Node Discovery & BGP

- Agent discovers nodes details and sync with other clusters
- Auto-generates FRRConfiguration

02. VTEP CIDR Allocation

- Allocates non overlapping CIDR

03. VNI/RT Assignment

- Dynamic allocation
- Auto sync across clusters

04. Stretch UDN

- "I want to stretch the blue CUDN across clusters"
- Create K8S CR (MCNC) pointing to CUDN to be stretched on each cluster

05. Stretch Default pod Net

- "I want to stretch pod Net across clusters"
- Create K8S CR (MCNC) with network type: default on each cluster

06. BGP/EVPN control

- Opinionated automation - simplified API but less control over BGP internals
- May not satisfy advanced networking requirements

Demo Env

OVN-K

Upstream OVN-K (April 22)

- With one change:
- Remove CUDN network immutability

Link to code:

<https://github.com/yboaron/ovn-kubernetes/tree/skynet-evpn-base>

Clusters

OVN-Kubernetes Kind

- 2 Clusters (cluster1, cluster2)
- Multi network enable, deploy-frr
- 1 Control + 2 Worker nodes each

DEPLOYMENT

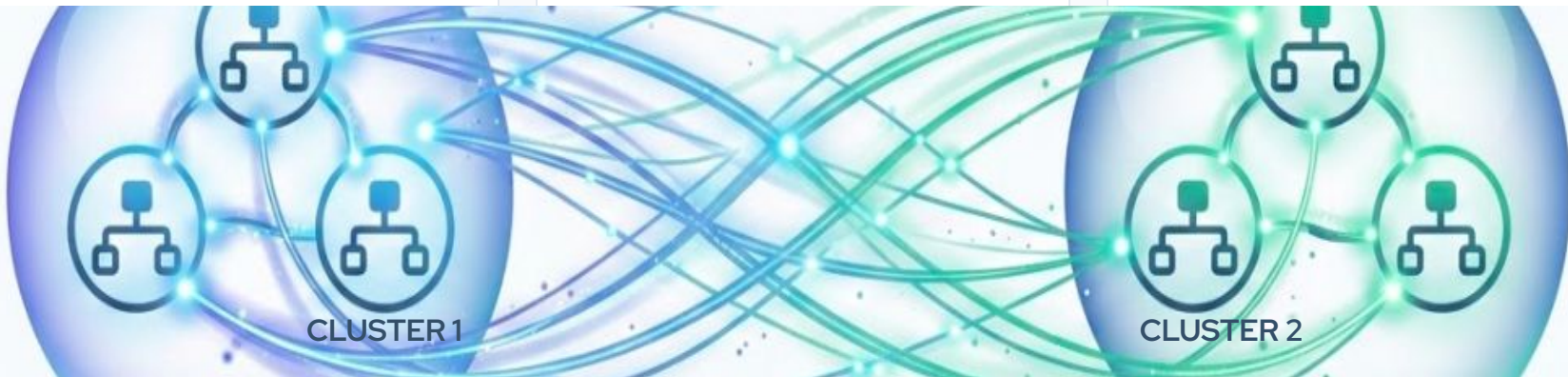
MCN

<https://github.com/yboaron/mcn/tree/default-network-connectivity>

"make deploy"

- Deploys 2 OVN-K Kind clusters + MCN
- VTEP in unmanaged mode
- **"make test-cudn-l3"** test cudn pod2pod traffic
- **"make test-default-network"**

Test default pod networking connectivity



Demo Recordings

BGP mesh

MCN Agent Demo: Infrastructure Setup

Goal: Show automated BGP peering using OVN-K APIs



CUDN stretch

MCN Agent Demo: CUDN Stretching with EVPN

Goal: Automate CUDN stretching using OVN-K APIs



Pod2Pod
across CUDN

MCN Agent Demo: Pod-to-Pod Connectivity

Goal: Verify EVPN datapath with VXLAN encapsulation



Pod2Pod

Default network

MCN Agent Demo: Default Network Connectivity

Goal: Extend default pod network across clusters via BGP



Thoughts?



Technical Considerations

Use Cases

What specific multicluster use cases should we solve?

A.2 Architecture

Should it be part of OVN-K core ?
some layer on top of OVN-K ?
Explore different architecture?



Next steps

B.1 Exploration

What other technical bits and integration points should be explored?

B.2 Formalization

Is it time to draft an OKEP to define requirements and design?